Networking Academy

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Lab - DNS Lookups

Objectives

Passive reconnaissance is a method of information gathering in which the tools do not interact directly with the target device or network. In this lab, you will explore common tools used to gather information about a target through the Domain Name System (DNS).

Use **nslookup** to obtain domain and IP address information.

Use the **whois** command to find additional registration information.

Compare the Output of the Nslookup and Dig tools.

Perform Reverse DNS Lookups.

Background / Scenario

Before beginning any penetration test or other ethical hacking engagement, you need to covertly obtain as much information about the target organization as possible. There is a wealth of information that can be obtained from publicly available domain registration data. In this lab, you will investigate the output of the **nslookup**, **whois**, and **dig** commands.

Required Resources

Kali VM customized for Ethical Hacker course

Internet access

Instructions

Part 1: Use nslookup to Obtain Domain and IP Address Information.

Step 1: Log into Kali Linux and access the terminal environment.

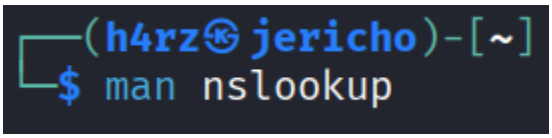
a. Log into the Kali system with the username **kali** and the password **kali**. You are presented with the Kali desktop.

b. Open a terminal window by clicking on the **Terminal** icon located near the top of the screen.

Step 2: Investigating nslookup capabilities

Nslookup is a command line tool that is available in Linux and Windows. Its basic usage is to convert a domain name to an IP address. Nslookup has other functionality that can provide additional information.

a. Access the manual pages for **nslookup** using the **man** command:



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b. To review the manual pages, press the **spacebar** to advance the pages. When you are finished reviewing the manual pages, press **q** to quit and return to the command line.

```
NSLOOKUP(1)                                BIND 9                                NSLOOKUP(1)

NAME
       nslookup - query Internet name servers interactively

SYNOPSIS
       nslookup [-option] [name | -] [server]

DESCRIPTION
       nslookup  is  a  program  to  query Internet domain name servers.  nslookup has two modes: interactive and
       non-interactive. Interactive mode allows the user to query name  servers  for  information  about  various
       hosts  and domains or to print a list of hosts in a domain.  Non-interactive mode prints just the name and
       requested information for a host or domain.

ARGUMENTS
       Interactive mode is entered in the following cases:

       a. when no arguments are given (the default name server is used);

       b. when the first argument is a hyphen (-) and the second argument is the host name or Internet address of
          a name server.

       Non-interactive mode is used when the name or Internet address of the host to be looked up is given as the
       first argument. The optional second argument specifies the host name or address of a name server.

       Options can also be specified on the command line if they precede the arguments and are  prefixed  with  a
       hyphen.  For  example, to change the default query type to host information, with an initial timeout of 10
       seconds, type:
 Manual page nslookup(1) line 1 (press h for help or q for help or q to quit)
```

```
querytype=value | type=value
       This keyword changes the type of the information query to value. The defaults are A and then
       AAAA; the abbreviations for these keywords are q and ty.

       Please note that it is only possible to specify one query type. Only  the  default  behavior
       looks up both when an alternative is not specified.
```

Which **set** keyword would you use to query for the mail server mx record within a domain?

set q=mx

```
┌──(h4rz㉿jericho)-[~]
└─$ nslookup
> set q=mx
> student.gmi.edu.my
Server:         192.168.88.2
Address:        192.168.88.2#53

Non-authoritative answer:
student.gmi.edu.my      mail exchanger = 1 aspmx.l.google.com.
student.gmi.edu.my      mail exchanger = 10 aspmx2.googlemail.com.
student.gmi.edu.my      mail exchanger = 10 aspmx3.googlemail.com.
student.gmi.edu.my      mail exchanger = 5 alt1.aspmx.l.google.com.
student.gmi.edu.my      mail exchanger = 5 alt2.aspmx.l.google.com.

Authoritative answers can be found from:
aspmx.l.google.com      internet address = 74.125.130.26
aspmx.l.google.com      has AAAA address 2404:6800:4003:c20::1b
aspmx3.googlemail.com   internet address = 172.217.78.26
aspmx3.googlemail.com   has AAAA address 2607:f8b0:4023:1c05::1b
alt1.aspmx.l.google.com internet address = 192.178.163.26
alt1.aspmx.l.google.com has AAAA address 2607:f8b0:400e:c17::1a
alt2.aspmx.l.google.com has AAAA address 2607:f8b0:4023:1c05::1b
>
```

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Step 3: Using the nslookup command

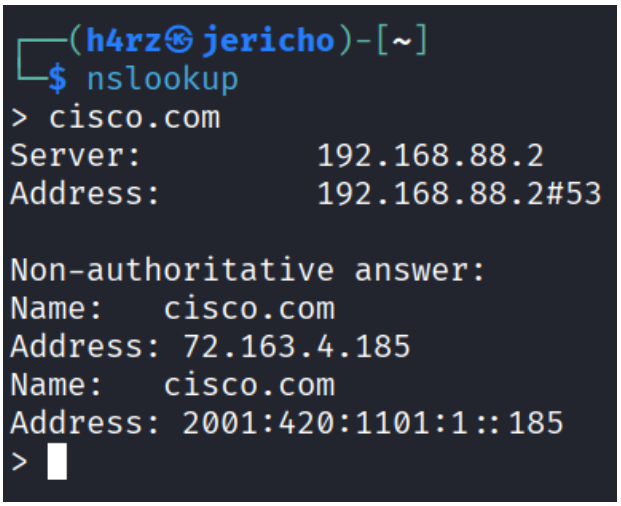
a. Use the **nslookup** command with no options to enter interactive mode. To exit interactive mode at any time, type **exit** to return to the CLI prompt.



b. The CLI prompt changes to > to indicate that you are now in interactive mode and can enter the various nslookup commands. Enter the domain name **cisco.com** to resolve the domain name to an IP address. By default, the **nslookup** command queries A and AAAA records for the target.

```
> cisco.com
```

The output of the command will be similar to that shown. The A record contains the IPv4 address assigned to the root domain and the AAAA record contains the IPv6 address.



c. To find the domain name servers configured for cisco.com, use the **set type** command to change the query type to “ns” to return the name server information.

```
> set type=ns
> cisco.com
```

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The output of the command should be similar to that shown below. The servers are listed by fully qualified domain name and are further listed as authoritative servers for both IPv4 and IPv6 addresses.

```
> set type=ns
> cisco.com
;; communications error to 192.168.1.1#53: timed out
Server:         192.168.1.1
Address:        192.168.1.1#53

Non-authoritative answer:
cisco.com       nameserver = ns1.cisco.com.
cisco.com       nameserver = ns3.cisco.com.
cisco.com       nameserver = ns2.cisco.com.

Authoritative answers can be found from:
ns2.cisco.com   internet address = 64.102.255.44
<output omitted>
```

What are the IPv4 and IPv6 addresses of the primary DNS server (ns1)?

IPv4 = 72.163.5.201

IPv6 = 2001:420:1101:6::a

By default IPv6 was not displayed. So we need to pull it with nslookup set to “set type=aaa” for network “ns1.cisco.com”.

d. Enter **exit** to leave interactive mode and return to the CLI prompt.

Example with gmi.edu.my.

```
┌──(h4rz㉿jericho)-[~]
└─$ nslookup
> set type=ns
> gmi.edu.my
Server:         192.168.88.2
Address:        192.168.88.2#53

Non-authoritative answer:
gmi.edu.my      nameserver = irena.ns.cloudflare.com.
gmi.edu.my      nameserver = apollo.ns.cloudflare.com.

Authoritative answers can be found from:
apollo.ns.cloudflare.com        internet address = 108.162.193.66
apollo.ns.cloudflare.com        internet address = 173.245.59.66
apollo.ns.cloudflare.com        internet address = 172.64.33.66
apollo.ns.cloudflare.com        has AAAA address 2606:4700:58::adf5:3b42
apollo.ns.cloudflare.com        has AAAA address 2a06:98c1:50::ac40:2142
apollo.ns.cloudflare.com        has AAAA address 2803:f800:50::6ca2:c142
irena.ns.cloudflare.com internet address = 108.162.194.79
irena.ns.cloudflare.com internet address = 172.64.34.79
irena.ns.cloudflare.com internet address = 162.159.38.79
irena.ns.cloudflare.com has AAAA address 2803:f800:50::6ca2:c24f
irena.ns.cloudflare.com has AAAA address 2a06:98c1:50::ac40:224f
irena.ns.cloudflare.com has AAAA address 2606:4700:50::a29f:264f
> exit


┌──(h4rz㉿jericho)-[~]
└─$
```

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e.

Step 4: Change the server used to perform lookups.

Occasionally it is desirable to use a different DNS server to perform lookups. This may be necessary if the local DNS server is unable to resolve an address or resolves the host name to an internal private address and you need to obtain the internet accessible address of the host.

a. In this query, use the one-line **nslookup** command syntax to change the server to look up skillsforall.com. The syntax for the command is **nslookup [hostname] [server IP].**

```
┌──(h4rz㉿jericho)-[~]
└─$ nslookup skillsforall.com 8.8.8.8
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   skillsforall.com
Address: 18.67.181.76
Name:   skillsforall.com
Address: 18.67.181.115
Name:   skillsforall.com
Address: 18.67.181.75
Name:   skillsforall.com
Address: 18.67.181.77
```

In interactive mode, you change the server using the **server** keyword.

```
┌──(h4rz㉿jericho)-[~]
└─$ nslookup
> server 8.8.8.8
Default server: 8.8.8.8
Address: 8.8.8.8#53
> skillsforall.com
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   skillsforall.com
Address: 18.67.181.76
Name:   skillsforall.com
Address: 18.67.181.77
Name:   skillsforall.com
Address: 18.67.181.75
Name:   skillsforall.com
Address: 18.67.181.115
>
```

b. The ***any*** query type can retrieve much, or all, of the information contained in the DNS record for a host name. Often ***text*** records that can provide additional details about the domain are contained in DNS records. Using the 8.8.8.8 Google DNS server, find the DNS records for skillsforall.com.

```
┌──(kali㉿Kali)-[~]
└─$ nslookup
> server 8.8.8.8
> set type=any
> skillsforall.com
```

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The output should look similar to this example:

```
┌──(kali㉿Kali)-[~]
└─$ nslookup
> server 8.8.8.8
Default server: 8.8.8.8
Address: 8.8.8.8#53
> set type=any
> skillsforall.com
;; Connection to 8.8.8.8#53(8.8.8.8) for skillsforall.com failed: timed out.
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   skillsforall.com
Address: 13.225.142.127
Name:   skillsforall.com
Address: 13.225.142.7
Name:   skillsforall.com
Address: 13.225.142.73
Name:   skillsforall.com
Address: 13.225.142.9
skillsforall.com        nameserver = ns-1130.awsdns-13.org.
skillsforall.com        nameserver = ns-1652.awsdns-14.co.uk.
skillsforall.com        nameserver = ns-489.awsdns-61.com.
skillsforall.com        nameserver = ns-588.awsdns-09.net.
skillsforall.com
        origin = ns-1130.awsdns-13.org
        mail addr = awsdns-hostmaster.amazon.com
        serial = 1
        refresh = 7200
        retry = 900
        expire = 1209600
        minimum = 86400
skillsforall.com        mail exchanger = 10 inbound-smtp.us-east-1.amazonaws.com.
skillsforall.com        text = "d1g1l9y74sxj8m.cloudfront.net"
skillsforall.com        text =
"facebook-domain-verification=8cg08gu4eikp0d2d1quqhjwh5ti1vv"
skillsforall.com        text =
"google-site-verification=Q5NIWRygJYTSLxuHReNKw1kvgC8IXKTOyPf5zITDv40"
skillsforall.com        text =
"identrust_validate=tadDBgWwQAKpw6QCCQDCagqsZgxHELybnPOCQHNU+rsV"
```

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My output :

```
┌──(h4rz㉿jericho)-[~]
└─$ nslookup
> server 8.8.8.8
Default server: 8.8.8.8
Address: 8.8.8.8#53
> set type=any
> skillsforall.com
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   skillsforall.com
Address: 13.35.238.95
Name:   skillsforall.com
Address: 13.35.238.70
Name:   skillsforall.com
Address: 13.35.238.20
Name:   skillsforall.com
Address: 13.35.238.67
skillsforall.com        nameserver = ns-588.awsdns-09.net.
skillsforall.com        nameserver = ns-1130.awsdns-13.org.
skillsforall.com        nameserver = ns-489.awsdns-61.com.
skillsforall.com        nameserver = ns-1652.awsdns-14.co.uk.
skillsforall.com
        origin = ns-1130.awsdns-13.org
        mail addr = awsdns-hostmaster.amazon.com
        serial = 1
        refresh = 7200
        retry = 900
        expire = 1209600
        minimum = 86400
skillsforall.com        mail exchanger = 10 inbound-smtp.us-east-1.amazonaws.com.
skillsforall.com        text = "facebook-domain-verification=8cg08gu4eikp0d2d1quqhjwh5ti1vv"
skillsforall.com        text = "v=spf1 include:amazonses.com ~all"
skillsforall.com        text = "identrust_validate=XzTu3rqoVVwnwNykPpaGYBeA4de5HaSynIEnsHWXyIur"
skillsforall.com        text = "google-site-verification=Q5NIWRygJYTSLxuHReNKw1kvgC8IXKTOyPf5zITDv40"
skillsforall.com        text = "d1g1l9y74sxj8m.cloudfront.net"

Authoritative answers can be found from:
>
```

What record types are displayed in the output of the nslookup command with the type set to any?

The `nslookup` output with type set to any displays A (IPv4 addresses), NS (Name Servers), SOA (Start of Authority), MX (Mail Exchanger), and TXT (Text) record types.

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Part 2: Use the whois command to find additional registration information.

The whois tool queries domain registration information, rather than the DNS server records. It is another form of passive reconnaissance that can identify where the domain is registered, technical and administrative contact information, and physical locations. Be aware that information contained in domain registrations can be set to private and often the contact information is that of the hosting service, rather than the organization itself.

Step 1: Compare whois output for various organizations.

a. The whois tool is available from the CLI prompt ion Kali Linux. Use the **whois** command to obtain information about cisco.com.



b. Now use the **whois** command to obtain information about the skillsforall.com domain.

What conclusion can you make about the two domains (cisco.com and skillsforall.com) based on the output of the **whois** commands?

Based on the matching registrant organization, address, and contact email in the WHOIS records, it can be concluded that both domains are owned and managed by Cisco Technology Inc.

Step 2: Use whois to determine IP address registration information.

The whois tool can also be used to gather information about IP address ranges that are assigned to an organization. In the previous part of this lab, we discovered the IP addresses assigned to various domain DNS server host names. Now you can use that address information to obtain additional details about the external IP address ranges that are assigned to those organizations.

a. Review the output you obtained from using **nslookup** to obtain the DNS server IP addresses for cisco.com. Record the IP addresses of the Cisco DNS servers.

b. Use the Whois tool to find what IP address ranges are assigned to Cisco and are used on the networks hosting their DNS servers. At the time of this lab, ns1.cisco.com resolved to the IP address 72.163.5.201, however this may vary. At the prompt, enter **whois 72.163.5.201**.

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```
┌──(h4rz㉿jericho)-[~]
└─$ whois 72.163.5.201

#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#


NetRange:       72.163.0.0 - 72.163.255.255
CIDR:           72.163.0.0/16
NetName:        CISCO-GEN-7
NetHandle:      NET-72-163-0-0-1
Parent:         NET72 (NET-72-0-0-0-0)
NetType:        Direct Allocation
OriginAS:
Organization:   CISCO SYSTEMS, INC. (CS-2831)
RegDate:        2006-10-24
Updated:        2024-03-08
Comment:        Geofeed https://www.cisco.com/web/automation/CiscoAS109_geoip.csv
Ref:            https://rdap.arin.net/registry/ip/72.163.0.0


OrgName:        CISCO SYSTEMS, INC.
OrgId:          CS-2831
Address:        251 LITTLE FALLS DRIVE
City:           WILMINGTON
StateProv:      DE
PostalCode:     19808
Country:        US
RegDate:        2023-06-07
Updated:        2024-02-16
Ref:            https://rdap.arin.net/registry/entity/CS-2831


OrgAbuseHandle: CISCO6-ARIN
OrgAbuseName:   Cisco CSIRT
OrgAbusePhone:  +1-408-527-3227
OrgAbuseEmail:  csirt-notify@cisco.com
OrgAbuseRef:    https://rdap.arin.net/registry/entity/CISCO6-ARIN

OrgNOCHandle: GATC-ARIN
OrgNOCName:   GIS ARIN Technical Contact
OrgNOCPhone:  +1-408-526-8888
OrgNOCEmail:  cidr-block-admins@cisco.com
OrgNOCRef:    https://rdap.arin.net/registry/entity/GATC-ARIN

OrgTechHandle: GATC-ARIN
OrgTechName:   GIS ARIN Technical Contact
OrgTechPhone:  +1-408-526-8888
OrgTechEmail:  cidr-block-admins@cisco.com
OrgTechRef:    https://rdap.arin.net/registry/entity/GATC-ARIN


#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
```

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What is the IP address range for the IPv4 addresses allocated to Cisco? The ns1.cisco.com server is addressed within this block.

72.163.0.0 - 72.163.255.255

c. Because organizations may use the same IP networks for other externally facing servers, knowing the address ranges is valuable for determining which networks to target during a penetration test. Use the whois tool to obtain the IP address allocations for the IP networks where the other Cisco DNS servers are located.

Part 3: Compare the Output of the nslookup and dig Tools

Step 1: Use Linux dig to Query for DNS servers.

a. Dig is a Linux tool that performs DNS queries. The format of a dig query is similar to nslookup. To resolve the hostname cisco.com to an IP address, use the syntax **dig [*hostname*].**

```
┌──(h4rz㉿jericho)-[~]
└─$ dig cisco.com

; <<>> DiG 9.20.15-2-Debian <<>> cisco.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 30974
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 512
;; QUESTION SECTION:
;cisco.com.                     IN      A

;; ANSWER SECTION:
cisco.com.              5       IN      A       72.163.4.185

;; Query time: 26 msec
;; SERVER: 192.168.88.2#53(192.168.88.2) (UDP)
;; WHEN: Mon Jan 26 21:33:06 EST 2026
;; MSG SIZE  rcvd: 54
```

What is the difference between the default record types queried by Dig and those queried by nslookup?

b. To obtain the IPv6 address of cisco.com it is necessary to add a type to the command structure. The syntax to instruct Dig to query a specific record type is **dig *[*hostname*] [*record type*]***.

```
┌──(h4rz㉿jericho)-[~]
└─$ dig cisco.com AAAA

; <<>> DiG 9.20.15-2-Debian <<>> cisco.com AAAA
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 65069
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 512
;; QUESTION SECTION:
;cisco.com.                     IN      AAAA

;; ANSWER SECTION:
cisco.com.              5       IN      AAAA    2001:420:1101:1::185

;; Query time: 11 msec
;; SERVER: 192.168.88.2#53(192.168.88.2) (UDP)
;; WHEN: Mon Jan 26 21:33:53 EST 2026
;; MSG SIZE  rcvd: 66
```

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Step 2: Use dig to Obtain Additional Information.

a. In the earlier part of this lab, nslookup was used to obtain the DNS servers for cisco.com. Use the 8.8.8.8 Google DNS server to query for the DNS server records. The syntax to use a dig command to perform a query using a different DNS server is **dig [*hostname*] @[*DNS server IP*] [*type*]**. At the prompt, enter **dig cisco.com 8.8.8.8 ns**.

```
┌──(h4rz㉿jericho)-[~]
└─$ dig cisco.com 8.8.8.8 ns

; <<>> DiG 9.20.15-2-Debian <<>> cisco.com 8.8.8.8 ns
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 23054
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;cisco.com.                     IN      A

;; ANSWER SECTION:
cisco.com.              5       IN      A       72.163.4.185

;; Query time: 3 msec
;; SERVER: 192.168.88.2#53(192.168.88.2) (UDP)
;; WHEN: Mon Jan 26 21:34:48 EST 2026
;; MSG SIZE  rcvd: 43

;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 15628
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 512
;; QUESTION SECTION:
;8.8.8.8.                       IN      NS

;; AUTHORITY SECTION:
.                       5       IN      SOA     a.root-servers.net. nstld.verisign-grs.com. 2026012602 1800 900 60
4800 86400

;; Query time: 7 msec
;; SERVER: 192.168.88.2#53(192.168.88.2) (UDP)
;; WHEN: Mon Jan 26 21:34:48 EST 2026
;; MSG SIZE  rcvd: 111
```

b. Earlier, nslookup was used with the ***set type=any*** option to find additional information about the skillsforall.com hostname. The ***any*** record type can also be queried using dig.

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```
┌──(h4rz㉿jericho)-[~]
└─$ dig @8.8.8.8 skillsforall.com any

; <<>> DiG 9.20.15-2-Debian <<>> @8.8.8.8 skillsforall.com any
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 41981
;; flags: qr rd ra; QUERY: 1, ANSWER: 15, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;skillsforall.com.              IN      ANY

;; ANSWER SECTION:
skillsforall.com.       60      IN      A       13.35.238.20
skillsforall.com.       60      IN      A       13.35.238.70
skillsforall.com.       60      IN      A       13.35.238.95
skillsforall.com.       60      IN      A       13.35.238.67
skillsforall.com.       21600   IN      NS      ns-588.awsdns-09.net.
skillsforall.com.       21600   IN      NS      ns-1652.awsdns-14.co.uk.
skillsforall.com.       21600   IN      NS      ns-1130.awsdns-13.org.
skillsforall.com.       21600   IN      NS      ns-489.awsdns-61.com.
skillsforall.com.       900     IN      SOA     ns-1130.awsdns-13.org. awsdns-hostmaster.amazon.com. 1 720
0 900 1209600 86400
skillsforall.com.       1800    IN      MX      10 inbound-smtp.us-east-1.amazonaws.com.
skillsforall.com.       1800    IN      TXT     "identrust_validate=XzTu3rqoVVwnwNykPpaGYBeA4de5HaSynIEnsH
WXyIur"
skillsforall.com.       1800    IN      TXT     "facebook-domain-verification=8cg08gu4eikp0d2d1quqhjwh5ti1
vv"
skillsforall.com.       1800    IN      TXT     "google-site-verification=Q5NIWRygJYTSLxuHReNKw1kvgC8IXKTO
yPf5zITDv40"
skillsforall.com.       1800    IN      TXT     "d1g1l9y74sxj8m.cloudfront.net"
skillsforall.com.       1800    IN      TXT     "v=spf1 include:amazonses.com ~all"

;; Query time: 23 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (TCP)
;; WHEN: Mon Jan 26 21:38:51 EST 2026
;; MSG SIZE  rcvd: 673
```

Compare the output of the dig tool with the output of nslookup for ***any*** record type. Which output is easier to read to obtain the values contained in the various record types?

The `dig` tool is generally easier to read because it provides a clean, tabular "Answer Section" that clearly aligns the record name, TTL, type, and value without the verbose, non-standardized labels found in `nslookup`.

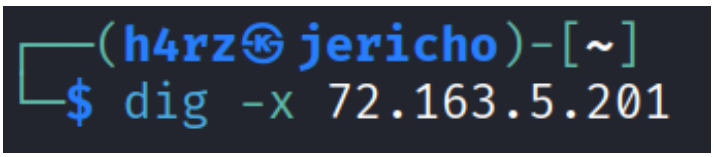
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Part 4: Perform Reverse DNS Lookups

Step 1: Use dig to Perform rDNS Lookups.

Now that you can perform DNS lookups and use Whois to determine IP address ranges, use dig to find additional host names. Reverse DNS (rDNS) lookups use the IP address to query for the host names of the services that resolve to that address.

a. Enter the **dig** command using the -x option to retrieve the hostname and record type of the ns1.cisco.com DNS server (**72.163.5.201)**.



What type of record is returned with the host name?

```
┌──(h4rz㉿jericho)-[~]
└─$ dig -x 72.163.5.201

; <<>> DiG 9.20.15-2-Debian <<>> -x 72.163.5.201
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 7439
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;201.5.163.72.in-addr.arpa.     IN      PTR

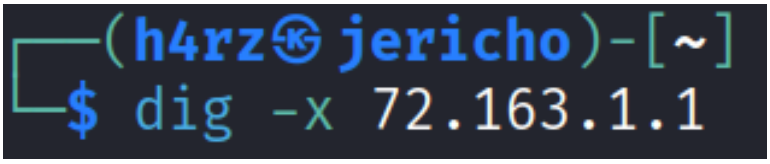
;; ANSWER SECTION:
201.5.163.72.in-addr.arpa. 5    IN      PTR     201.5.163.72.in-addr.arpa.

;; Query time: 3 msec
;; SERVER: 192.168.88.2#53(192.168.88.2) (UDP)
;; WHEN: Mon Jan 26 21:48:27 EST 2026
;; MSG SIZE  rcvd: 82
```

The type of record returned with the host name is a PTR (Pointer) record.

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b. Use the **dig -x** command to query for another IP address in the same subnet.



Examine the output returned from the dig command. What type of device do you think is assigned the 72.163.1.1 address?

```
┌──(h4rz㉿jericho)-[~]
└─$ dig -x 72.163.1.1

; <<>> DiG 9.20.15-2-Debian <<>> -x 72.163.1.1
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 50974
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0×0005, udp: 512
;; QUESTION SECTION:
;1.1.163.72.in-addr.arpa.       IN      PTR

;; ANSWER SECTION:
1.1.163.72.in-addr.arpa. 5      IN      PTR     hsrp-72-163-1-1.cisco.com.

;; Query time: 243 msec
;; SERVER: 192.168.88.2#53(192.168.88.2) (UDP)
;; WHEN: Mon Jan 26 21:45:39 EST 2026
;; MSG SIZE  rcvd: 91
```

Based on the PTR record, this address is assigned to a Cisco networking device (likely a router or Layer 3 switch) acting as a redundant default gateway using Hot Standby Router Protocol (HSRP).

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Step 2: Use the Host Utility to Perform rDNS Lookups.

The Host utility is a tool in Linux that performs lookups to convert IP addresses to host names. Use this utility to find another host on the 72.163.0.0/16 network.

a. The syntax of the **host** command is **host** [*ip address or hostname*]

```
┌──(h4rz㉿jericho)-[~]
└─$ host 72.163.10.1
Host 1.10.163.72.in-addr.arpa. not found: 3(NXDOMAIN)
```

b. Host can also be used to perform a quick IP address lookup for a known hostname.

```
┌──(h4rz㉿jericho)-[~]
└─$ host hsrp-72-163-10-1.cisco.com
Host hsrp-72-163-10-1.cisco.com not found: 3(NXDOMAIN)
```

How does the output of the host command differ from dig or nslookup when querying for an IP address assigned to a known host?

Unlike the verbose, metadata-rich output of `dig` or `nslookup`, the `host` command returns a concise, single-line summary that simply states the domain name pointer associated with the IP address.

c. URLs often contain aliases for the host name of the server hosting the website. The output of the host command can list the servers that respond to that URL.

```
┌──(h4rz㉿jericho)-[~]
└─$ host www.cisco.com
www.cisco.com is an alias for www.cisco.com.akadns.net.
www.cisco.com.akadns.net is an alias for wwwds.cisco.com.edgekey.net.
wwwds.cisco.com.edgekey.net is an alias for wwwds.cisco.com.edgekey.net.globalredir.akadns.net.
wwwds.cisco.com.edgekey.net.globalredir.akadns.net is an alias for e2867.dsca.akamaiedge.net.
e2867.dsca.akamaiedge.net has address 23.220.240.118
e2867.dsca.akamaiedge.net has IPv6 address 2001:e68:2:b8::b33
e2867.dsca.akamaiedge.net has IPv6 address 2001:e68:2:95::b33
```

The information about aliases is useful when trying to determine where the actual website or service is located.

Step 3: Use nslookup to Perform rDNS Lookups

Nslookup is used primarily to perform IP address lookups for known host names. It can also be used to perform rDNS lookups to return a host name assigned to a known IP address.

Use Nslookup to find hostnames associated with an IP address.

In non-interactive mode the syntax to do an rDNS query is nslookup [ip address].

```
┌──(h4rz㉿jericho)-[~]
└─$ nslookup 72.163.5.201
201.5.163.72.in-addr.arpa       name = ns1.cisco.com.
```

To use interactive mode, enter **nslookup** with no options. At the **>** prompt, enter the target IP address.

```
┌──(h4rz㉿jericho)-[~]
└─$ nslookup
> 72.163.5.201
201.5.163.72.in-addr.arpa       name = 201.5.163.72.in-addr.arpa.
```

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Reflection

In this lab, you used nslookup, dig, and host to obtain information from DNS zone files. Which tool would you use to begin a passive reconnaissance effort against a targeted domain? Why?

I would use `whois` to begin passive reconnaissance because it identifies the target's organizational ownership, physical locations, and registered IP address ranges without ever interacting directly with the target's own servers.

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